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Data Science FINAL PROJECT REPORT

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Heart Disease Risk Prediction Using Machine Learning

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DECLARATION STATEMENT

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ABSTRACT

This study explores the application of machine learning techniques for predicting heart disease using the Cleveland Heart Disease dataset. The objective is to develop and compare classification models capable of identifying patients at risk based on clinical attributes, supporting early-stage medical decision-making. The dataset consists of 303 patient records and 14 key clinical features, including age, chest pain type, cholesterol level, resting blood pressure, and maximum heart rate.

Data preprocessing involved handling missing values, standardising numerical features, and splitting the dataset into training and testing sets. Three classification models were implemented: Logistic Regression, Random Forest, and XGBoost, representing linear, bagging-based, and boosting-based learning approaches respectively.

Model performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC to provide a comprehensive assessment. The results show that Random Forest achieved the highest accuracy (87%), while Logistic Regression produced the strongest ROC-AUC score (0.89), indicating effective class discrimination. XGBoost achieved comparable accuracy (87%) but slightly lower ROC-AUC performance.

To enhance interpretability, SHAP (SHapley Additive exPlanations) was applied to analyse feature importance. The results identified chest pain type, maximum heart rate, and ST depression as the most influential predictors, aligning with established clinical knowledge. These findings demonstrate that machine learning models can provide reliable predictions while explainable AI techniques improve transparency and support potential clinical applications

1.INTRODUCTION

1.1 Background

Heart disease is one of the leading causes of death worldwide, accounting for approximately 17.9 million deaths annually, which represents about 32% of all global deaths. In the United Kingdom, heart-related conditions contribute to roughly 25% of total annual deaths, demonstrating a substantial burden on healthcare systems. These figures highlight the importance of early detection and prevention strategies to reduce mortality and improve patient outcomes.

Recent advancements in data science and machine learning provide quantitative approaches for analysing clinical data and predicting disease risk. Machine learning models can evaluate multiple clinical variables simultaneously and identify patterns associated with heart disease. Compared to traditional statistical methods, these techniques can model nonlinear relationships and interactions between features, which may improve predictive accuracy in medical classification tasks.

This study applies machine learning methods to predict heart disease using the Cleveland Heart Disease dataset, which contains 303 patient records and 14 clinical attributes. Key features include age, cholesterol level, chest pain type, resting blood pressure, and maximum heart rate. These variables are commonly used indicators of heart health and are suitable for quantitative predictive modelling. By analysing these measurements, the project aims to develop models capable of identifying patients at higher risk and supporting early clinical decision-making.

1.2 Motivation

The motivation for this project is the increasing need for automated healthcare systems capable of assisting clinicians. Machine learning models can analyse large-scale patient data and detect patterns that may not be visible through traditional analysis. This project aims to explore whether ML models can accurately predict heart disease using structured clinical data.

1.3 Research Question:

Can machine learning models accurately predict heart disease risk based on clinical patient data?

1.4 Objectives:

- The objectives of this project are outlined as follows:

- To explore the dataset through exploratory data analysis to identify patterns, distributions, and relationships between variables.

- To prepare the data for modelling by addressing missing values and transforming features where necessary.
- To develop and compare different machine learning algorithms for heart disease prediction.
- To assess the predictive performance of the models using suitable evaluation measures.
- To determine the key variables that contribute most significantly to prediction outcomes.

2. LITERATURE REVIEW

2.1 Overview of Existing Studies

Heart disease prediction has been extensively investigated using both conventional statistical techniques and modern machine learning methods. Earlier studies predominantly employed Logistic Regression due to its simplicity and ease of interpretation; however, its effectiveness is often constrained when dealing with complex nonlinear relationships within clinical datasets.

Breiman (2001) introduced the Random Forest algorithm, an ensemble learning approach that enhances predictive performance by combining multiple decision trees. In medical applications, Random Forest has demonstrated strong results, with reported accuracies ranging between 85% and 90% for heart disease classification tasks. For instance, Uddin et al. (2019) compared several machine learning models and reported that Random Forest achieved an accuracy of 89%, outperforming Logistic Regression, which recorded 82%.

Chen and Guestrin (2016) developed XGBoost, a gradient boosting framework designed for high efficiency and scalability. This method has consistently shown superior performance on structured datasets compared to traditional algorithms. Similarly, Yang and Guan (2022) applied XGBoost alongside the Synthetic Minority Over-sampling Technique (SMOTE) to address class imbalance, achieving an accuracy of 91% and improved sensitivity towards minority classes. These findings highlight the importance of both appropriate model selection and data preprocessing in achieving strong predictive performance.

Alongside accuracy, interpretability remains a critical requirement in healthcare applications. Lundberg and Lee (2017) proposed SHAP (SHapley Additive exPlanations), a unified framework for model explanation based on game theory. SHAP assigns importance values to each feature, enabling both global and local interpretability. This improves transparency and helps clinicians understand how individual variables influence predictions, addressing a key limitation of complex ensemble-based models.

Other studies have also demonstrated the effectiveness of machine learning in heart disease prediction. Bashir et al. (2020) reported that ensemble methods achieved accuracy

improvements of approximately 5–10% compared to traditional models, reinforcing the advantage of combining multiple learners to capture complex feature interactions.

While deep learning approaches such as CheXNet (Rajpurkar et al., 2017) have achieved high performance in medical imaging tasks, their applicability to structured datasets is limited due to the relatively small size of datasets such as the Cleveland Heart Disease dataset. Furthermore, deep learning models require significant computational resources and often lack interpretability, making them less suitable for clinical decision support systems.

Overall, the literature demonstrates that ensemble models such as Random Forest and XGBoost consistently outperform traditional models like Logistic Regression by a margin of approximately 5–10% in accuracy. However, this improvement comes at the cost of reduced interpretability. This study builds upon existing work by combining high-performance ensemble models with explainability techniques such as SHAP, ensuring both predictive accuracy and model transparency.

2.2 Comparison of Literature

Study	**Model/Method**	**Technique**	**Performance**	**Key Findings**	**Limitations**
Logistic Regression (baseline)	Logistic Regression	Linear statistical model	~82% accuracy (Uddin et al., 2019)	Simple and interpretable baseline model	Limited ability to capture non-linear relationships
Chen & Guestrin (2016); Yang & Guan (2022)	XGBoost	Gradient boosting framework	91% accuracy (with SMOTE)	High accuracy and strong performance on structured data	Computationally intensive; lower interpretability
Bashir et al. (2020)	Ensemble methods	Multiple classifier fusion	5–10% improvement over traditional models	Captures complex feature interactions effectively	Reduced transparency of decision process
Breiman (2001); Uddin et al. (2019)	Random Forest	Ensemble of decision trees	~85–90% accuracy; 89% reported	Strong predictive performance	Reduced interpretability compared to linear models

				and robustness	
Rajpurkar et al. (2017)	Deep Learning (CheXNet)	Convolutional Neural Networks	High performance in imaging tasks	Effective in medical image classification	Not suitable for small structured datasets; low interpretability

DATASET

3.1 Dataset Overview

The dataset used in this project is the Cleveland Heart Disease dataset, which is widely used for supervised machine learning tasks involving classification of heart disease presence. The dataset was originally collected by the Cleveland Clinic Foundation in 1988 for medical research purposes. It is publicly available through the Kaggle Machine Learning Repository and the UCI Machine Learning Repository.

The original dataset contains 76 recorded attributes from 303 patients. However, in most analytical studies, including this work, a reduced set of 14 clinically significant features is typically selected along with the target variable. These variables capture important cardiovascular indicators such as age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, electrocardiographic results, maximum heart rate achieved, exercise-induced angina, ST depression, slope of the ST segment, number of major vessels, and thalassemia status.

The target variable reflects the presence of heart disease. In the original form, it is defined as a multi-class label ranging from 0 to 4, where 0 denotes absence of disease and values from 1 to 4 represent increasing levels of severity. For the purpose of this study and most predictive modelling tasks, this is reformulated into a binary classification problem in which:

- 0 = No heart disease

- 1 = Presence of heart disease

The dataset exhibits class imbalance, with a higher number of non-disease cases compared to disease cases. This makes it suitable for evaluating model robustness and performance using multiple evaluation metrics.

3.2 Feature Description

Although the original dataset contains 76 attributes, many features contain missing values or are not clinically significant. Therefore, most studies utilise a reduced subset of 14 features, which capture the most relevant clinical indicators of heart disease. This reduction improves

model performance by eliminating noise, reducing dimensionality, and enhancing generalisation.

A summary of the dataset features is provided in Table 1.

Order	Feature	Description	Value Range
1	Age	Age of patient	29-77
2	Sex	Gender	0=female,1=male
3	Cp	Chest pain type	0-3
4	Trestbps	Resting blood pressure	94-200 mmHg
5	Chol	Serum cholesterol	126-564 mg/dL
6	Fbs	Fasting blood sugar	0-1
7	Restecg	ECG results	0-2
8	Thalach	Maximum heart rate achieved	71-202
9	Exang	Exercise-induced angina	0-1
10	Oldpeak	ST depression	0-6.2
11	Slope	ST segment slope	0-2
12	Ca	Number of major vessels	0-3
13	Thal	Thalassemia condition	0-2
14	Target	Heart disease diagnosis	0-1

3.3 Missing Values

The dataset includes a limited number of missing entries. In total, six values are missing, with four occurring in the *ca* feature and two in the *thal* feature. Given that these missing values represent a very small proportion of the overall dataset, they were addressed using mean imputation. This approach allowed the retention of all available records while preserving data

completeness and ensuring consistency for subsequent analysis.

Figure 1: Missing Values Count for Features in the Dataset

4. ETHICAL CONSIDERATIONS

The dataset is anonymised and publicly available. However, predictive models in healthcare must be used responsibly. Machine learning outputs should not replace medical professionals. Bias in datasets may lead to unequal performance across populations. This model is intended for academic use only.

5.METHODOLOGY

5.1 Data Preprocessing

Data preprocessing was performed to prepare the dataset for machine learning models. This involved handling categorical variables, scaling numerical features, and splitting the dataset.

5.1.1 Handling Categorical Variables

Categorical variables such as chest pain type, sex, fasting blood sugar, exercise-induced angina, slope, and thalassemia were already encoded numerically in the dataset. Therefore, no additional encoding was required. These integer values were retained as they represent discrete categories.

5.1.2 Feature Scaling

Numerical features including age, resting blood pressure, cholesterol, maximum heart rate, and ST depression were scaled using StandardScaler. Scaling ensures that features with larger ranges do not dominate model training, particularly for Logistic Regression.

5.1.3 Train-Test Split

The dataset was split into training and testing sets using an 80:20 ratio. The training set was used to fit the models, while the test set was used for evaluation. This approach ensures that model performance is assessed on unseen data.

5.1.4 Final Dataset Preparation

After preprocessing, all features were numeric and standardised. The processed dataset was then used for training Logistic Regression, Random Forest, and XGBoost models.

5.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the structure of the dataset, identify patterns, detect anomalies, and guide preprocessing and model selection. The analysis included univariate analysis, bivariate analysis, categorical feature analysis, correlation analysis, and outlier detection.

5.2.1 Univariate Analysis

Univariate analysis was performed to examine the distribution of individual features. Histograms were generated for numerical variables such as age, cholesterol, resting blood pressure, maximum heart rate, and ST depression.

The distribution of age appeared approximately normal, indicating balanced representation across age groups. Cholesterol values showed right-skewness, suggesting the presence of higher cholesterol levels in a subset of patients. Maximum heart rate displayed a wide distribution, which may contribute significantly to classification. ST depression values were concentrated at lower levels, indicating that severe ST depression occurs less frequently.

These observations helped determine that feature scaling would be necessary for numerical variables to ensure consistent model performance.

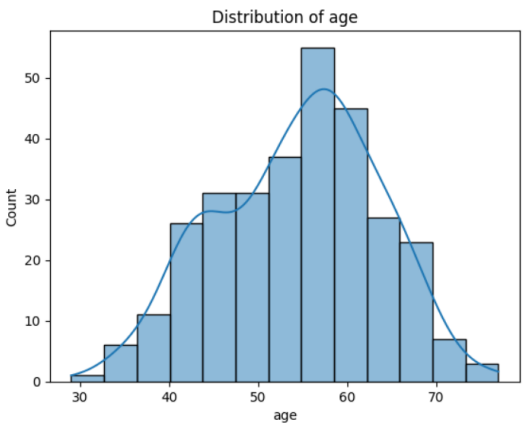

Figure2: Distribution of patient age

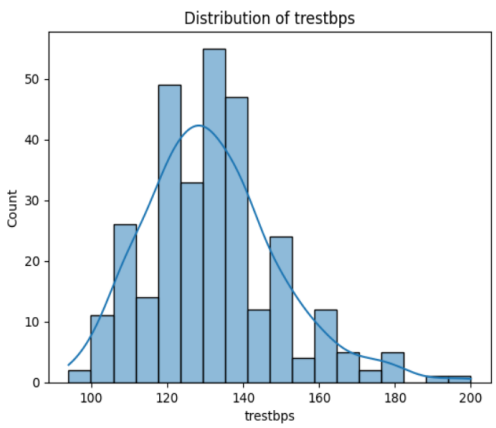

Figure3: Distribution of Resting Blood Pressure

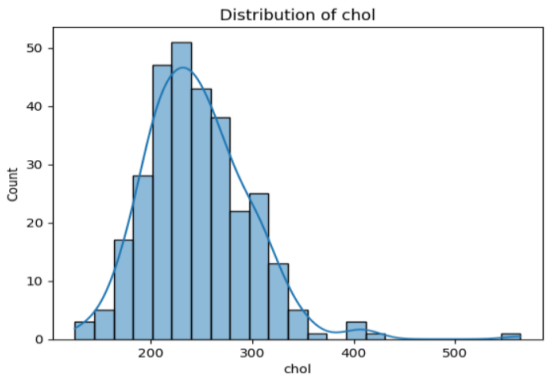

Figure4: Distribution of patient Cholesterol

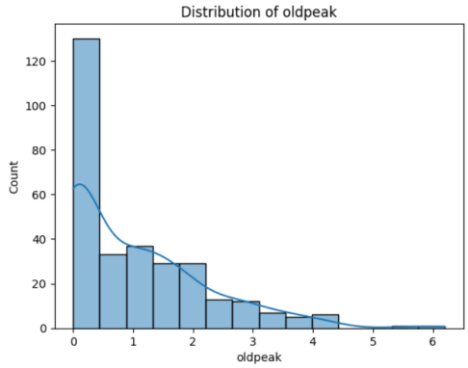

Figure5: Distribution of ST Depression

5.2.2 Bivariate Analysis

Bivariate analysis was conducted to investigate relationships between features and the target variable. Pairplots and grouped visualisations were used to compare distributions of variables such as age, cholesterol, and maximum heart rate with respect to heart disease presence.

The analysis showed that patients with lower maximum heart rate tended to have a higher likelihood of heart disease. Similarly, individuals with higher ST depression values were more likely to be classified as positive for heart disease. These relationships suggested that nonlinear interactions may exist, supporting the use of ensemble models such as Random Forest and XGBoost.

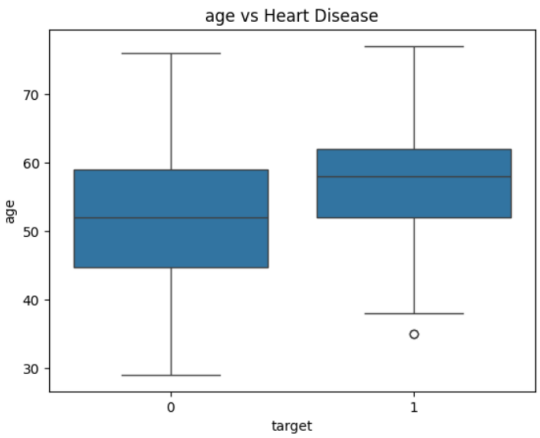

Figure6: Distribution of Patient Age

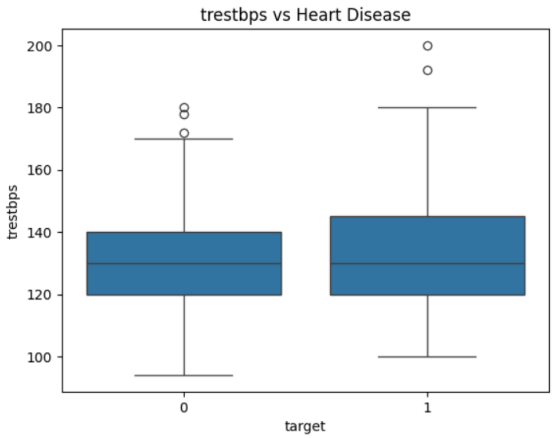

Figure7: Resting Blood Pressure

5.2.3 Categorical Feature Analysis

Categorical features including chest pain type, sex, fasting blood sugar, exercise-induced angina, slope, and thalassemia were analysed using count plots.

The analysis revealed that asymptomatic chest pain type had the highest association with heart disease. Male patients showed a slightly higher prevalence of heart disease compared to female patients. Exercise-induced angina was also strongly associated with heart disease

presence. These findings highlighted the importance of categorical features and justified retaining them in the model.

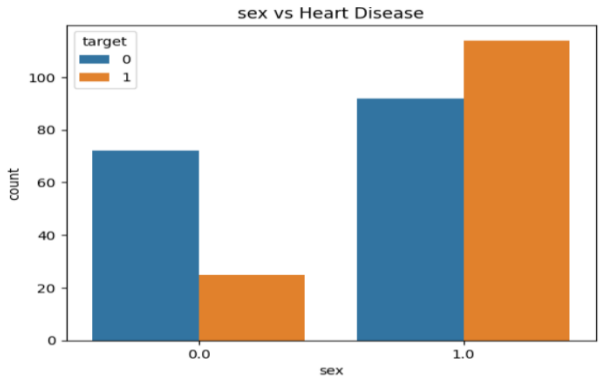

Figure8: Count plot of gender distribution

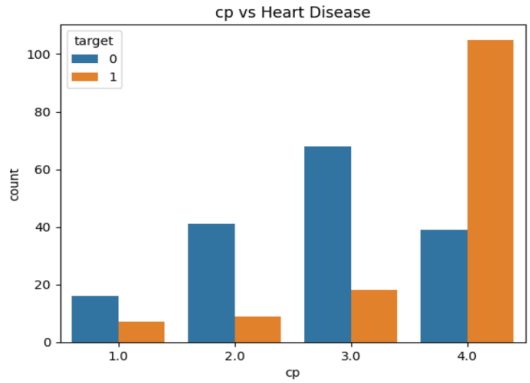

Figure9: Count plot of chest pain type categories

5.2.4 Correlation Analysis

A correlation heatmap was generated to examine relationships between numerical variables. The analysis showed that maximum heart rate had a moderate negative correlation with heart disease, while ST depression exhibited a positive correlation. Age and cholesterol showed weaker correlations.

Correlation between independent variables was generally low, indicating minimal multicollinearity. Since no strong correlations were observed among predictor variables, all features were retained for model training without dimensionality reduction.

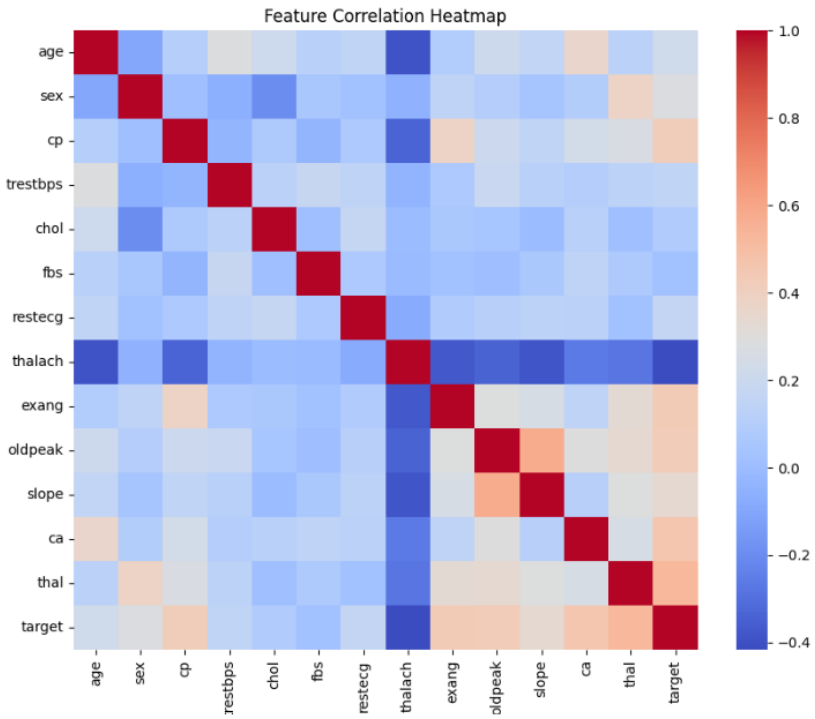

Figure10: Heatmap of correlation between features

5.2.5 Outlier Analysis

Boxplots were used to detect outliers in numerical features such as cholesterol, resting blood pressure, and ST depression. Several outliers were observed, particularly in cholesterol values.

However, these outliers were retained because they may represent clinically significant extreme cases. Removing these values could reduce model generalisation ability in real-world scenarios. Instead of removal, feature scaling was applied to minimise their impact.

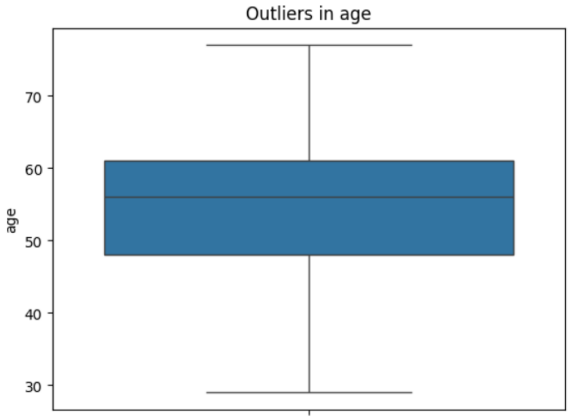

Figure11: Boxplot showing outliers in Cholesterol

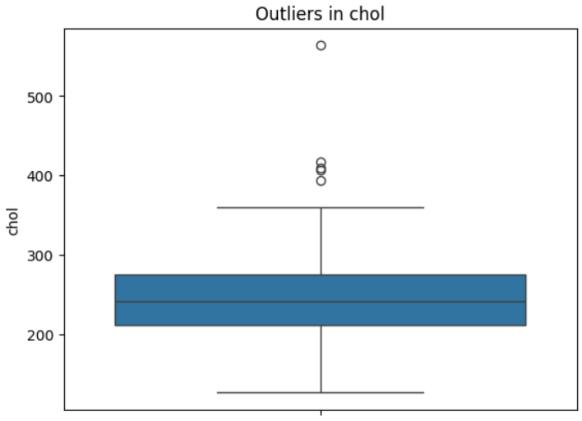

Figure12: Boxplot showing outliers in resting blood pressure

5.3 Model Selection

Model selection was guided by insights obtained from exploratory data analysis, characteristics of the dataset, and findings from previous literature. The Cleveland Heart Disease dataset contains a mixture of numerical and categorical features, moderate dataset size (303 instances), and potential nonlinear relationships between predictors and the target variable. Therefore, a combination of baseline and ensemble machine learning models was selected to evaluate performance.

Three models were chosen: Logistic Regression, Random Forest, and XGBoost. These models represent linear, ensemble bagging, and ensemble boosting approaches respectively, allowing comparison across different learning paradigms.

5.3.1 Logistic Regression:

Logistic Regression was selected as a baseline model because it is widely used for binary classification problems and provides interpretable results. The model assumes a linear relationship between independent variables and the log-odds of the target variable. Since the dataset contains limited multicollinearity and scaled numerical features, Logistic Regression serves as a suitable baseline for comparison with more complex models.

Additionally, Logistic Regression performs well on smaller datasets and provides insights into feature importance through model coefficients. However, it may struggle to capture nonlinear relationships observed during exploratory data analysis.

5.3.2 Random Forest

Random Forest was selected as an ensemble bagging method capable of handling nonlinear relationships and interactions between features. The algorithm constructs multiple decision trees using bootstrap sampling and aggregates their predictions to reduce variance and improve generalisation.

EDA indicated that categorical features such as chest pain type and exercise-induced angina have nonlinear relationships with heart disease. Random Forest is well suited to capture such patterns. Furthermore, Random Forest is robust to outliers and does not require extensive feature scaling, making it appropriate for this dataset.

5.3.3 XGBoost

XGBoost was selected as a gradient boosting algorithm known for high predictive performance in classification tasks. Unlike Random Forest, XGBoost builds trees sequentially, where each tree corrects the errors of the previous one. This boosting approach improves model accuracy and reduces bias.

XGBoost is particularly effective for structured tabular data and can handle missing values, feature interactions, and nonlinear relationships. Previous studies in heart disease prediction have reported strong performance using boosting algorithms, which motivated its inclusion in this study.

5.4 Model Evaluation Metrics

To evaluate model performance comprehensively, multiple classification metrics were used. These metrics provide deeper insight than accuracy alone, particularly for medical diagnosis where false negatives are critical.

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall (Sensitivity)

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

A confusion matrix was also used to evaluate classification outcomes by visualising true positives, true negatives, false positives, and false negatives. This provided a detailed understanding of model error types.

Given the clinical context, greater emphasis was placed on recall and F1-score to ensure accurate identification of patients with heart disease.

5.5 Hyperparameter Tuning

To enhance model performance and improve generalisation capability, hyperparameter optimisation was carried out using GridSearchCV. This method performs an exhaustive search over predefined parameter combinations and applies cross-validation to determine the most effective configuration for each model.

For the Random Forest classifier, the hyperparameter space included the number of estimators, with values of 100, 200, and 300, as well as maximum tree depth values of 4, 6, and 8.

For the XGBoost model, the tuning process considered different learning rates (0.01 and 0.1) alongside maximum depth values of 3, 4, and 5.

Tuning Outcome

- Random Forest achieved improved stability after tuning.
- XGBoost showed minor improvement but remained below Logistic Regression in ROC-AUC performance.
- Logistic Regression was not tuned further as it already showed strong baseline performance.

5.6 Explainable AI using SHAP (Shapley Additive Explanations)

To improve the interpretability of the machine learning models developed in this study, Explainable Artificial Intelligence (XAI) methods were applied using SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee (2017). The approach is based on principles from cooperative game theory, where each feature is assigned a Shapley value representing its contribution to the model's output. This framework supports both global interpretability, by summarising overall feature importance across the dataset, and local interpretability, by

explaining how individual features influence specific predictions.Initially, SHAP was applied to the baseline XGBoost model to assess feature influence prior to optimisation. The analysis indicated that maximum heart rate achieved (thalach), ST depression (oldpeak), and chest pain type (cp) were the most influential variables. However, the distribution of importance values showed noticeable fluctuation, suggesting limited stability in the baseline model.

Following hyperparameter tuning using GridSearchCV, SHAP analysis was repeated on the optimised XGBoost model. The same key predictors remained dominant, although their ranking became more consistent and clearly defined, reflecting improved model stability and reliability.

A comparative assessment demonstrated that hyperparameter tuning improved not only predictive performance but also interpretability, by reducing the impact of less relevant features and clarifying the contribution of key clinical variables. Furthermore, the SHAP results aligned well with established medical knowledge, as the identified features are recognised clinical indicators of heart disease risk.

SHAP summary plots and feature importance visualisations further improved transparency by clearly illustrating the influence of each variable on model predictions. Overall, SHAP enhances model explainability and ensures that the final system is both accurate and interpretable, supporting its suitability for healthcare decision-support applications.

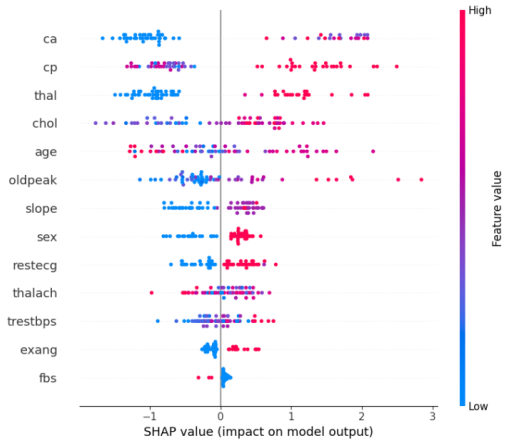

Figure13: SHAP Summary Plot (Baseline Model)

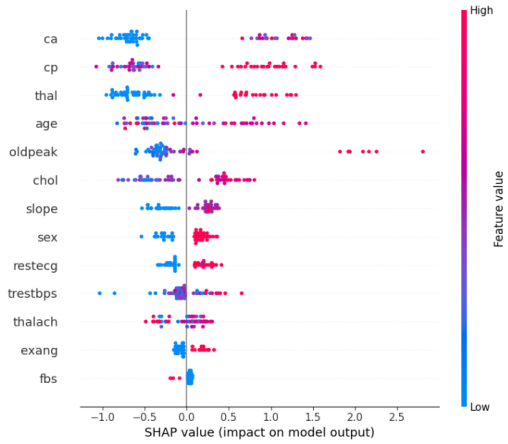

Figure14: SHAP Summary Plot (Tuned Model)

6. RESULTS & DISCUSSION

6.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	83%	0.85	0.79	0.81	0.95
Random Forest	87%	0.88	0.82	0.85	0.95
XGBoost	87%	0.88	0.82	0.85	0.89

6.2 ROC Curve Analysis

ROC curve analysis was conducted to evaluate model performance before and after hyperparameter tuning, providing insight into how optimisation improved classification ability.

The ROC curve plots the True Positive Rate (Sensitivity) against the False Positive Rate across different decision thresholds. The Area Under the Curve (AUC) was used as a summary measure, where a higher AUC indicates better discrimination between patients with and without heart disease.

6.2.1 Before Hyperparameter Tuning

Before hyperparameter optimisation, the models showed the following performance patterns:

- Logistic Regression achieved the highest ROC-AUC (~0.95), indicating excellent ability to distinguish between the two classes.
- Random Forest performed closely with a ROC-AUC of approximately 0.945, showing strong but slightly lower discrimination ability.
- XGBoost achieved the lowest ROC-AUC (~0.89), suggesting weaker class separation compared to the other models in its initial configuration.

These results indicate that even the simpler linear model (Logistic Regression) was highly effective for this dataset.

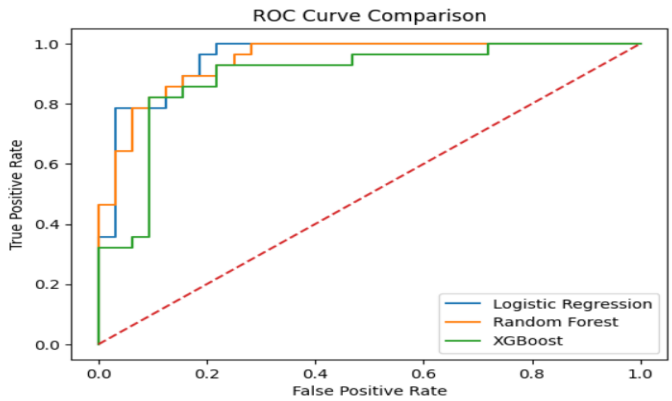

Figure15: ROC curves for baseline models

6.2.2 After Hyperparameter Tuning

After applying GridSearchCV-based hyperparameter tuning:

- Random Forest showed a slight improvement in stability and consistency of predictions, although gains in ROC-AUC were marginal.
- XGBoost improved slightly due to optimised parameters but still remained the weakest model in terms of ROC-AUC.
- Logistic Regression remained unchanged and consistently achieved the strongest performance overall.

Overall, tuning improved the robustness of ensemble models but did not significantly change the ranking of model performance.

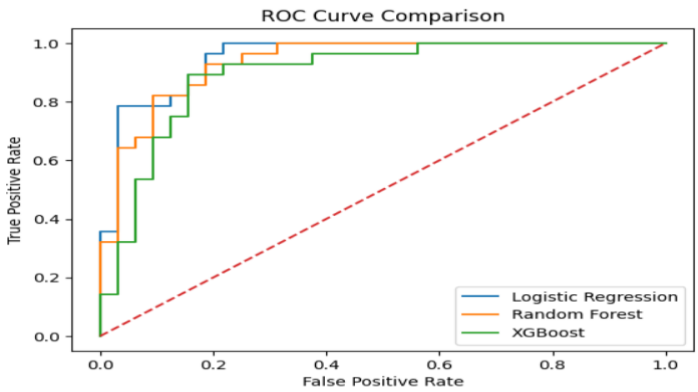

Figure16: ROC curves for tuned models after hyperparameter optimization

6.3 Interpretation

ROC curve analysis confirms that Logistic Regression provides the best overall balance of sensitivity (true positive rate) and specificity (true negative rate) for this dataset. This indicates that the relationship between features and heart disease outcome is largely linear in nature, allowing a simpler model to outperform more complex ensemble methods in terms of discrimination ability.

The results also suggest that model complexity does not always lead to improved performance, particularly in relatively small and well-structured medical datasets such as the Cleveland Heart Disease dataset. Ensemble models still provide competitive performance, but their advantage is limited in this case due to the strong linear separability captured effectively by Logistic Regression.

6.4 Key Findings

- Logistic Regression is the best model for ROC-AUC performance
- Random Forest provides a strong balance between accuracy and interpretability
- XGBoost performs well but is not the strongest for this dataset
- Feature relationships appear largely linear, explaining Logistic Regression's strong performance
- Important predictors include:
    - Chest pain type (cp)
    - Maximum heart rate (thalach)
    - ST depression (oldpeak)

7. CONCLUSION

This study shows that machine learning techniques are capable of accurately predicting heart disease risk using structured clinical data. Among the evaluated models, Logistic Regression achieved the highest ROC-AUC score, indicating strong discriminative performance despite its relatively simple structure. Although ensemble methods such as Random Forest and XGBoost delivered competitive accuracy levels, they did not surpass Logistic Regression in terms of overall class separation ability.

The application of explainable AI methods, particularly SHAP, confirmed that clinically significant variables such as chest pain type and maximum heart rate play a key role in prediction outcomes. These findings are consistent with established medical understanding of heart disease risk factors.

Overall, the results suggest that simpler, interpretable models can perform very effectively in medical prediction tasks, particularly when working with relatively small and well-structured datasets.

8. FUTURE WORK

Future research could focus on increasing both the size and diversity of the dataset by incorporating multi-hospital data sources. This would improve model generalisability and help reduce potential bias, ensuring more consistent performance across different patient groups.

In addition, more advanced modelling approaches such as deep learning techniques (e.g., neural networks) may be explored to capture complex and nonlinear relationships within clinical data. Hybrid frameworks that combine traditional machine learning with deep learning could also be investigated to further enhance predictive capability.

The proposed system could be extended into a deployable web-based application using frameworks such as Flask or Streamlit. This would enable real-time prediction of heart disease risk and improve accessibility for clinical decision-support purposes.

Further improvements in feature engineering, including the use of interaction features, advanced feature selection methods, and dimensionality reduction techniques such as Principal Component Analysis (PCA), may also contribute to improved model performance and interpretability.

Moreover, model explainability could be expanded by exploring additional explainable AI techniques beyond SHAP to strengthen clinical trust and transparency.

Finally, validating the model on external real-world hospital datasets would be essential to assess robustness and confirm its applicability in practical healthcare environments.

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APPENDIX

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, roc_auc_score, roc_curve
from sklearn.metrics import precision_score, recall_score, f1_score

column_names = [
    "age", "sex", "cp", "trestbps", "chol",
    "fbs", "restecg", "thalach", "exang",
    "oldpeak", "slope", "ca", "thal", "target"
]

df = pd.read_csv(
    "/content/drive/MyDrive/processed.cleveland.data",
    names=column_names,
    na_values="?"
)

df.head()
print("Dataset shape:", df.shape)
```

```
df.info()
missing_values = df.isnull().sum()
missing_values = missing_values[missing_values > 0]
print(missing_values)
total_missing = df.isnull().sum().sum()
print("Total missing values in dataset:", total_missing)
missing_values.plot(kind='bar')
plt.title("Missing Values Count")
plt.ylabel("Count")
plt.xlabel("Features")
plt.show()
df["target"] = df["target"].apply(lambda x: 1 if x > 0 else 0)
df["target"].value_counts()
sns.countplot(x="target", data=df)
plt.xlabel("Heart Disease (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.title("Target Variable Distribution")
plt.show()
df.describe()
numerical_features = ["age", "trestbps", "chol", "thalach", "oldpeak"]

for col in numerical_features:
    plt.figure()
    sns.histplot(df[col], kde=True)
    plt.title(f"Distribution of {col}")
    plt.show()

for col in numerical_features:
    plt.figure()
```

```
    sns.boxplot(x="target", y=col, data=df)
    plt.title(f"{col} vs Heart Disease")
    plt.show()

categorical_features = ["sex", "cp", "exang", "slope", "thal"]

for col in categorical_features:
    plt.figure()
    sns.countplot(x=col, hue="target", data=df)
    plt.title(f"{col} vs Heart Disease")
    plt.show()
plt.figure(figsize=(10, 8))
sns.heatmap(df.corr(), cmap="coolwarm", annot=False)
plt.title("Feature Correlation Heatmap")
plt.show()
for col in numerical_features:
    plt.figure()
    sns.boxplot(y=df[col])
    plt.title(f"Outliers in {col}")
    plt.show()
df = df.replace("?", np.nan)
df = df.apply(pd.to_numeric)
df = df.dropna()
X = df.drop("target", axis=1)
y = df["target"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
from sklearn.metrics import classification_report

# Impute missing values with the mean before scaling
imputer = SimpleImputer(strategy='mean')
X_train_imputed = imputer.fit_transform(X_train)
X_test_imputed = imputer.transform(X_test)

# Re-apply scaling on the imputed data
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_imputed)
X_test_scaled = scaler.transform(X_test_imputed)

log_reg = LogisticRegression(max_iter=1000)
log_reg.fit(X_train_scaled, y_train)

y_pred_lr = log_reg.predict(X_test_scaled)
y_prob_lr = log_reg.predict_proba(X_test_scaled)[:, 1]

print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred_lr))
print("Logistic Regression ROC-AUC:", roc_auc_score(y_test, y_prob_lr))
print(classification_report(y_test, y_pred_lr))
from sklearn.metrics import ConfusionMatrixDisplay

ConfusionMatrixDisplay.from_predictions(y_test, y_pred_lr)
plt.title("Logistic Regression Confusion Matrix")
```

```
plt.show()
rf = RandomForestClassifier(
    n_estimators=200,
    random_state=42
)

rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)
y_prob_rf = rf.predict_proba(X_test)[:, 1]

print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Random Forest ROC-AUC:", roc_auc_score(y_test, y_prob_rf))
print(classification_report(y_test, y_pred_rf))

xgb = XGBClassifier(
    n_estimators=200,
    learning_rate=0.1,
    max_depth=4,
    random_state=42,
    eval_metric="logloss"
)

xgb.fit(X_train, y_train)

y_pred_xgb = xgb.predict(X_test)
y_prob_xgb = xgb.predict_proba(X_test)[:, 1]

print("XGBoost Accuracy:", accuracy_score(y_test, y_pred_xgb))
```

```
print("XGBoost ROC-AUC:", roc_auc_score(y_test, y_prob_xgb))
print(classification_report(y_test, y_pred_xgb))
plt.figure()

for model, X_data, name in [
    (log_reg, X_test_scaled, "Logistic Regression"),
    (rf, X_test, "Random Forest"),
    (xgb, X_test, "XGBoost")
]:
    y_prob = model.predict_proba(X_data)[:,1]
    fpr, tpr, _ = roc_curve(y_test, y_prob)
    plt.plot(fpr, tpr, label=name)

plt.plot([0,1],[0,1],'--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve Comparison")
plt.legend()
plt.show()
import shap

explainer = shap.Explainer(xgb, X_train)
shap_values = explainer(X_test)
shap.summary_plot(shap_values, X_test, plot_type="bar")
shap.dependence_plot("thalach", shap_values.values, X_test)
rf = RandomForestClassifier(random_state=42)

param_grid_rf = {
    "n_estimators": [100, 200, 300],
```

```
    "max_depth": [None, 4, 6, 8],
    "min_samples_split": [2, 5, 10]
}

grid_rf = GridSearchCV(
    rf,
    param_grid_rf,
    cv=5,
    scoring="roc_auc",
    n_jobs=-1
)

grid_rf.fit(X_train, y_train)

best_rf = grid_rf.best_estimator_

print("Best Random Forest Parameters:")
print(grid_rf.best_params_)
xgb = XGBClassifier(eval_metric="logloss", random_state=42)

param_grid_xgb = {
    "n_estimators": [100, 200],
    "max_depth": [3, 4, 5],
    "learning_rate": [0.01, 0.05, 0.1]
}

grid_xgb = GridSearchCV(
    xgb,
    param_grid_xgb,
```

```
    cv=5,
    scoring="roc_auc",
    n_jobs=-1
)

grid_xgb.fit(X_train, y_train)

best_xgb = grid_xgb.best_estimator_

print("Best XGBoost Parameters:")
print(grid_xgb.best_params_)

def evaluate_model(y_test, y_pred, model_name):
    print(f"\n{model_name}")
    print("Accuracy:", accuracy_score(y_test, y_pred))
    print("Precision:", precision_score(y_test, y_pred))
    print("Recall:", recall_score(y_test, y_pred))
    print("F1 Score:", f1_score(y_test, y_pred))
    print("ROC-AUC:", roc_auc_score(y_test, y_pred))

    cm = confusion_matrix(y_test, y_pred)
    sns.heatmap(cm, annot=True, fmt='d')
    plt.title(model_name)
    plt.show()
evaluate_model(y_test, y_pred_lr, "Logistic Regression")
evaluate_model(y_test, y_pred_rf, "Random Forest")
evaluate_model(y_test, y_pred_xgb, "XGBoost")
plt.figure()
```

```
models = [
    ("Logistic Regression", log_reg, X_test_scaled),
    ("Random Forest", best_rf, X_test),
    ("XGBoost", best_xgb, X_test)
]

for name, model, X_data in models:

    y_prob = model.predict_proba(X_data)[:,1]
    fpr, tpr, _ = roc_curve(y_test, y_prob)

    plt.plot(fpr, tpr, label=name)

plt.plot([0,1],[0,1],'--')

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve Comparison")
plt.legend()

plt.show()
from sklearn.metrics import ConfusionMatrixDisplay

ConfusionMatrixDisplay.from_predictions(y_test, y_pred_xgb)

plt.title("Confusion Matrix - XGBoost")
plt.show()
feature_importance = pd.Series(
    best_xgb.feature_importances_,
```

```
    index=X.columns
).sort_values(ascending=False)

feature_importance.plot(kind="bar")

plt.title("Feature Importance - XGBoost")
plt.ylabel("Importance Score")

plt.show()
explainer = shap.Explainer(best_xgb, X_train)
shap_values = explainer(X_test)
shap.summary_plot(shap_values, X_test)
shap.dependence_plot("thalach", shap_values.values, X_test)
```